PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

umhackathon@um.edu.my

TABLE OF CONTENTS

1. Project Overview	**3**
2. Background & Business Objective	**4**
3. Product Purpose	**5**
4. System Functionalities	**6**
5. User Stories & Use Cases	**8**
6. Features Included (Scope Definition)	**11**
7. Features Not Included (Scope Control)	**12**
8. Assumptions & Constraints	**13**
9. Risks & Questions Throughout Development	**15**

Project Name: Smezy

Version: 1.0

1. Project Overview

- **Problem Statement:**
  SME owners often struggle to make financial decisions due to a lack of clear visibility into their cash flow situation. They rely on manual calculations or assumptions to decide whether to delay payments, collect invoices or take loans. Without a proper system to simulate outcomes, these decisions can lead to cash shortages or poor financial planning.

- **Target Domain:**
  Supporting SME financial management and decision-making by providing tools that help users understand, predict and evaluate financial situations before taking action.

- **Proposed Solution Summary:**
  Smezy helps SME owners simulate financial decisions before executing them. Users can input financial data and test different scenarios such as delaying payments, collecting invoices, or taking loans. The system processes this input, generates financial outcomes, and provides clear explanations and recommendations. This allows users to make better decisions based on measurable impact and understanding.

2. Background & Business Objective

- **Background of the Problem:**
  Many SME owners manage their financial decisions using manual methods such as spreadsheets or personal judgement. The process is often not structured, and decisions are made without clear visibility of future cash flow. Actions like delaying payments, collecting invoices, or taking loans are done without understanding their impact, which can lead to financial instability or cash shortages.

- **Importance of Solving This Issue:**
  Without proper decision support, SMEs are at higher risk of making poor financial choices that can affect business survival. Providing a system that can simulate financial outcomes helps users understand the consequences of their decisions before taking action. This reduces uncertainty and improves financial planning.

- **Strategic Fit / Impact:**
  The system supports the use of AI in business decision-making by integrating financial simulation with AI-generated insights. It aligns with the growing adoption of AI tools in helping businesses make faster and more informed decisions. The system also supports SME digital transformation by moving from manual financial management to a more structured and intelligent approach.

3. Product Purpose

3.1. Main Goal of the System:

To provide an AI-powered decision support system that helps SME owners simulate financial decisions and understand their impact before taking action.

3.2. Intended Users (Target Audience):

- SME Owners
- Entrepreneurs
- Small Business Manager
- Financial Staff in SMEs
- Business Consultants

4. System Functionalities

4.1. Description:

The system operates as an AI-driven financial decision support tool that allows SME users to simulate different financial scenarios before taking action. It follows a structured flow where user input is processed through a simulation engine and then interpreted by an AI model to generate explanations, recommendations, and measurable impact.

4.2. Key Functionalities:

- **What-If Decision Simulator:**
  Allows users to test financial decisions such as delaying payments, collecting invoices, or taking loans, and view the calculated outcome before execution.

- **Cash Flow Runway Predictor:**
  Estimates how long the business can sustain based on current cash flow and highlights the expected date of cash shortage.

- **AI Explanation & Recommendation:**
  Converts simulation results into clear explanations and suggests the most suitable financial decision.

- **Risk & Impact Analysis:**
  Shows the risk level and expected financial impact of each decision, including changes in cash position and short-term stability.

4.3. AI Model & Prompt Design

4.3.1. Model Selection:

The system uses a multi-layer LLM pipeline consisting of Google Gemini as the primary model, Z.AI GLM as a secondary fallback model, and a local rule-based system as the final fallback. This layered approach ensures system reliability even when external APIs fail. It is selected because it can generate structured explanations, interpret numerical data, and provide clear reasoning for financial decisions. This makes it suitable for converting simulation results into understandable insights for SME users.

4.3.2. Prompting Strategy:

The system uses a structured multi-step prompting approach. First, the simulation engine generates numerical results. Then, these results are combined with user input into a structured prompt before being sent to the model. This approach ensures the AI focuses on the given data and produces consistent and relevant explanations for each scenario.

4.3.3. Context & Input Handling:

The system limits the input size by only sending relevant financial data such as total cash, expenses, invoices, and selected scenarios. If the input exceeds the acceptable size, the system summarises the data before sending it to the model. Irrelevant or excessive details are removed to ensure the prompt remains within the model's token limit.

4.3.4. Fallback & Failure Behavior:

If the AI model returns an unclear or irrelevant response, the system will retry the request with a simplified prompt. If the issue continues, the system will display a default message with simulation results only, without AI explanation. This ensures the user still receives usable output even if the AI response fails.

5. User Stories & Use Cases

- **User Stories:**
  As an SME owner, the user wants to input financial data such as cash balance, expenses, and invoices so that they can understand their current financial situation clearly. The user also wants to test different financial decisions, such as delaying payments, collecting invoices, or taking a loan, in order to see the possible outcomes before making a real decision. In addition, the user expects the system to provide clear explanations and recommendations so that they can understand the reasoning behind each suggested action. The user also wants to know the risk level associated with each decision to avoid running out of cash. Lastly, the user wants to compare different scenarios so that they can choose the most suitable financial decision for their business.

- **Use Cases (Main Interactions):**

  - **What-If Decision Flow:**
    The user enters financial data such as cash balance, expenses, and invoices. The user then selects a scenario (e.g. delay payment, collect invoice, take loan). The system processes the input through the simulation engine and AI module, and then displays the result including cash impact, risk level, recommendation, and explanation.

  - **Cash Flow Prediction Flow:**
    The user inputs current financial data. The system calculates the cash flow runway and predicts how long the business can sustain. The result is displayed together with a suggested action to improve the situation.

- **AI Explanation Flow:**
  After the simulation is completed, the system sends the results to the AI model. The AI generates a clear explanation and recommendation. The system then displays this explanation together with the financial outcome for the user.

- **Use Case Diagram:**

6. Features Included (Scope Definition)

- AI-powered financial decision simulator for invoice payment scenarios and cash flow evaluation
- Cash flow runway prediction based on cash balance, revenue, expenses, and seasonal adjustment
- AI financial insight generation using structured prompts and LLM-based reasoning
- PDF invoice data extraction (vendor, amount, tax, due date, PO reference) from digital text-based PDFs
- Smart payment decision engine that classifies invoices into Pay Now, Negotiate Installments, or Delay Payment
- Risk classification system (low, medium, high) based on cash flow ratio analysis
- Decision explainability module with step-by-step reasoning chain for AI outputs
- Financial impact dashboard showing RM-based computation and remaining cash after decisions
- Multi-step prompting framework combining structured financial data with AI input
- Automated context handling to filter and optimize AI input data
- Graceful fallback mechanism using rule-based logic when AI model fails

7. Features Not Included (Scope Control)

- Live integration with banking APIs (e.g., Maybank, CIMB, real-time account syncing)
- Automated payment execution or real financial transaction processing
- OCR support for scanned or image-based invoices (only digital PDFs supported)
- Multi-currency accounting and foreign exchange ledger management
- Long-term data storage or database persistence across sessions (no SQL/NoSQL backend)
- Multi-user real-time collaboration system
- Production-grade deployment backend infrastructure

8. Assumptions & Constraints

- **LLM Cost Constraint:**

  The AI model is used to generate action recommendations, clear explanations, quantifiable impact, and dynamic risk assessment based on user scenarios. To control cost, the system limits the amount of data sent to the model and focuses only on essential inputs rather than full datasets.

  Output prompts are kept concise and structured by only limited to less than 40 words to reduce token usage. This approach ensures the system remains cost-efficient while still providing meaningful AI-driven insights.

- **Technical Constraints:**

  The system is developed as a prototype and does not include full backend integration such as database storage or live bank API connections. Financial data is manually input by users and is not synced with external systems.

  The document scanner is limited to processing digital PDF files with selectable text and only can read RM (ringgit Malaysia) currency. While it does not integrate with external accounting software or real-time financial services.

- **Performance Constraints:**

  The AI dashboard supports only simple outputs and cannot handle multiple complex prompts simultaneously. AI-generated recommendations may vary in accuracy. Where the system relies on external AI services such as Google Gemini and Z.AI, which may experience latency or temporary service interruptions.

  While input processing runs smoothly, the performance may slow down with larger or more complex data, affecting dashboard update speed.

- **User Input:**

  The system requires users to manually input financial data, such as cash balance, revenue, and expenses, for the cash flow predictor to generate accurate results and update the dashboard. Users may also upload financial documents through the scanner to extract relevant data.

  While the AI chatbox can operate independently, features involving financial simulation and decision analysis depend on user-provided data. Therefore, the system is designed as a decision-support tool and still requires human input before taking action.

9. Risks & Questions Throughout Development

- **Design Similarity:**
  AI-generated recommendations and explanations may be similar across different users due to limited prompt variation and structured outputs. The challenge is ensuring responses remain relevant and not overly generic.

- **Frame Stability:**
  There is a risk of inconsistent system behavior, such as delayed dashboard updates or failed AI responses due to model latency or service interruptions. This may affect the overall user experience, especially when the system is handling larger inputs or multiple requests.

- **Drawing Interpretation**:
  The AI may misinterpret financial inputs or user scenarios, especially when the data is incomplete or unclear. This may affect the accuracy and relevance of AI-generated outputs, including recommendations and risk assessments.